



PARTHA DHAR

Backend and platform engineer

Senior Software Developer, Technical Solution Architect remit since 2025

CONTACT

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SKILLS AND TOOLS

Languages and frameworks

Python PHP / Laravel
Node.js TypeScript
Bash Solidity

Cloud and delivery

AWS Docker
GitLab CI
GitHub Actions nginx
Apache systemd

Data

PostgreSQL MySQL
SQLite DuckDB Redis
MongoDB

Security and quality

SonarQube opengrep
grype trivy fail2ban
iptables / ipset

EDUCATION

BTech, Computer Science and Engineering

B. P. Poddar Institute of Management and Technology 2009 - 2012

Diploma, Computer Science and Technology

Bengal Institute of Technology, Katwa 2005 - 2008

AWARD

Excellence in Customer Service

Brainium Information Technologies, 2018-19

SUMMARY

Backend and platform engineer, twelve years in, mostly on production systems that cannot be taken down. Senior Software Developer at Brainium Information Technologies since May 2016, with a Technical Solution Architect remit since 2025: feasibility analysis, risk registers, stack recommendations, schema design and cloud cost estimation.

Built cryptocurrency exchange infrastructure handling real money, with BTC, ETH and TRON full nodes operated in production. Incrementally modernised a legacy AWS and ColdFusion estate whose database holds records back to 1994, with no staging environment to practise on. Owned a serverless invoice pipeline end to end (51 of 52 commits) and set up the team's self-hosted code-quality gate.

By late 2022, I had written the backend for 4 business applications, 11 Node.js and MongoDB microservices and 5 Web3 services, set up CI/CD and Docker automation for 2 organisations, and managed 15+ VPS, 2 AWS accounts and 1 dedicated server. Every audit I write ends with the limits of that audit: what I did not check, and which of my earlier conclusions turned out to be wrong.

EXPERIENCE

Senior Software Developer

May 2016 - Present

Brainium Information Technologies Pvt. Ltd., Kolkata. Technical Solution Architect remit since 2025.

- Operate a legacy AWS and ColdFusion estate whose Aurora MySQL database holds records back to 1994, with no staging environment and no test suite: staged deployment with a rollback path built on top of production, production access read-only by default, mail relayed through SES, Fail2Ban and iptables hardening. (2025 - present)
- Diagnosed a production incident on that estate in which an autonomous blocking tool merged two flagged addresses into a /16 and dropped a search crawler's entire published range, discarding hundreds of thousands of crawler packets while tens of thousands of verified crawler ranges sat unread one layer above. Implemented a mandatory interlock that tests range overlap before every kernel-affecting firewall write, and restricted the authoritative allowlist to feeds published by the crawler operators themselves. (2026)
- Sole author (51 of 52 commits) of a serverless invoice ingestion pipeline on AWS Lambda, SAM and EventBridge: format-aware UBL extraction, 48 gated migrations behind a snapshot, a read-only preflight and a verify step, keyset-cursor chunked ETL inside the 900 s Lambda limit, and upserts that never overwrite a human edit. 15,244 invoices re-transformed with zero errors; 21.9 GiB of duplicated JSON identified in a 26.2 GiB database. Live in production for its first provider. (2025 - 2026)
- Built the team's quality gate: self-hosted SonarQube Community with about 20 curated analyser plugins, GitLab CI pipelines that block on the gate, and a 402-line Python bridge filing findings back as labelled, de-duplicated GitLab issues, adopted into three client pipelines. (2023 - 2024)
- Second committer on three public interfaces of a Uniswap-V2-style multi-chain AMM across TRON, BitTorrent Chain and BSC: 146 commits and 8 merged pull requests from the organisation's first week, per-chain router and factory wiring and CREATE2 init-code hashes, deploy workflows to Vercel and Heroku. Live since 2022. (2022 - 2023)

LANGUAGES

Bengali (native)

Hindi (native)

English (professional working proficiency)

INTERESTS

Books, Temples, History

- Built cryptocurrency exchange infrastructure handling real money: installed and operated Bitcoin, Ethereum and TRON full nodes in production, Node.js bridge APIs for deposits and withdrawals to blockchain wallets, a WebSocket order-allocation server, and 2FA and JWT on the user side.

Earlier product work, 2016 to 2023: a cryptocurrency trading-bot platform across two generations, Laravel first and then Node.js with a Dockerised Python exchange connector, where I wrote the at-rest encryption module holding users' own exchange API keys and read subscription tier from an on-chain NFT balance; a token presale back office; a job portal issuing a wildcard subdomain per recruiter; a device trade-in integration over partner APIs and headless browsers; a gaming portal fed by WebSocket balance updates; and a ride-hailing backend on Socket.IO.

Laravel Node.js Python TypeScript / React MySQL / PostgreSQL / MongoDB / Redis
AWS (Lambda, SAM, RDS, SES) Docker GitLab CI / GitHub Actions SonarQube
Web3 / TRON / ETH nginx / Apache / systemd

Web Developer

Nov 2013 - May 2016

Digimason Web Pvt. Ltd., Kolkata

PHP developer on six CodeIgniter applications, one e-commerce delivery across Magento, PrestaShop and OpenCart, WordPress builds, and operational ownership of two high-traffic magazine sites: self-taught PHP compilation, Apache load sharing and nightly deployments, because the sites needed it.

CodeIgniter WordPress Magento / PrestaShop / OpenCart Apache MySQL

SELECTED PERSONAL WORK

- **listenbrainz-widget** - Self-hosted now-playing SVG cards for ListenBrainz, MIT. Display-privacy flags are bound to the token at issue time, so editing the URL cannot widen them. ETag plus a soft-refresh parameter to beat a proxy that cached the fallback art.
- **Audio_Quality_Profiling** - Audio library quality and duplicate scanner: Python, ffpprobe, mutagen. A lossy codec at a high sample rate is an upsample: flagged at 0.9 confidence, score halved. Every scoring path returns its evidence, not a bare number. Ran clean on a 4,100-file library.
- **wb-electoral-data** - Bulk text-recovery pipeline for deliberately garbled government PDFs. Text came out as shifted garbage until a character-identifier (CID) mapping was built to decode the embedded fonts. 7,936 PDFs downloaded and validated, extraction logged at about 7.6 s per PDF, cross-check API calls self-throttled to 2 req/s under a published 50 req/s limit.
- **Bengali audiobook pipeline** - Text-to-speech pipeline for a Bengali epic: 76 commits in 10 days, 7 h 24 min of narrated audio, about 210 hand-approved corrections, run under a 4-thread cap on a 6-vCPU VPS with no GPU.
- **Vehicle telemetry server** - OBD-II telemetry server with 714 tests on SQLite and MariaDB, built against a 26.7M-datapoint corpus. Measurement reversed the design: 0 of 25 recorded drives completed a 0-60 run.
- **Multi-scanner aggregation** - Patched an upstream multi-scanner aggregator (grype, trivy, snyk and osv-scanner into one SQLite store) with three measured fixes so it actually aggregated: shared OSV identifiers 0 to 92, diff exposures 45 to 93, 43.6% coverage. A single scanner reporting zero is not evidence of safety; the product is the set of CVEs they disagree about.
- **A/B harness for an agent skill** - Three scenarios, two graded and one recorded as skipped, three runs per arm, six written assertions per scenario. Reported result: null, with cost deltas inside run-to-run variance.